

RIVERSIDE, CA, USA

WMO: 722869

Lat: 33.952N

Lon: 117.439W

Elev: 245

StdP: 98.41

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 03171

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	2.6	3.8	-17.2	0.9	16.5	-14.1	1.1	18.0	11.9	16.9	10.5	17.3	1.1	60	0.398

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.2	38.2	20.6	36.8	20.3	35.0	19.7	22.7	34.2	21.9	33.3	21.2	32.2	4.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	14.2	26.8	17.9	13.2	25.9	17.3	12.8	25.2	67.7	34.3	65.0	33.3	62.3	32.5	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.2	6.1	0.1	42.1	1.3	2.1	-0.9	43.6	-1.6	44.8	-2.4	46.0	-3.4	47.6		
(5)			-2.0	23.9	1.0	0.8	-2.7	24.5	-3.3	24.9	-3.8	25.3	-4.5	25.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.0	13.5	13.5	15.4	17.0	19.6	22.6	26.0	26.2	24.7	20.5	16.2	12.9
	DBStd	5.74	3.53	3.54	3.42	3.41	2.98	2.87	2.67	2.63	3.34	3.41	3.54	3.10
	HDD10.0	26	8	6	2	1	0	0	0	0	0	0	1	9
	HDD18.3	754	154	140	104	65	19	2	0	0	1	16	82	171
	CDD10.0	3327	118	104	168	210	297	379	495	503	440	327	188	98
	CDD18.3	1013	5	6	12	24	58	131	237	244	191	84	19	2
	CDH23.3	11601	99	138	251	411	682	1370	2545	2687	2054	987	319	57
CDH26.7	5907	16	34	76	157	293	672	1407	1533	1132	468	111	6	
Wind	WSAvg	2.2	2.2	2.1	2.1	2.3	2.4	2.5	2.4	2.2	2.0	1.9	2.0	2.1
Precipitation	PrecAvg	294	60	75	40	18	6	2	4	2	5	11	25	44
	PrecMax	565	288	284	168	62	51	18	38	10	38	140	54	215
	PrecMin	103	2	2	0	1	0	0	1	0	0	0	6	4
	PrecStd	134	74	64	37	14	10	4	7	2	8	26	13	45
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.6	30.0	32.1	34.1	36.2	38.4	39.7	40.2	40.2	37.2	32.8	27.5
		MCWB	12.7	13.3	15.0	15.7	18.1	19.7	21.2	20.8	20.3	17.8	14.7	12.6
	2%	DB	26.2	27.4	28.7	31.0	32.7	35.1	37.3	37.8	37.7	34.3	29.9	25.0
		MCWB	11.9	12.6	14.0	15.0	17.3	19.6	21.4	21.0	20.3	16.2	13.9	11.4
	5%	DB	23.8	24.6	26.2	27.9	30.2	32.8	35.5	36.2	35.5	32.0	27.2	22.5
		MCWB	11.1	12.0	13.5	14.4	17.0	19.1	20.8	20.6	19.8	15.9	13.0	10.5
	10%	DB	21.6	21.3	23.4	25.2	27.3	30.9	33.7	34.1	33.0	29.0	24.1	20.0
		MCWB	10.6	10.9	12.9	13.9	16.6	18.7	20.4	20.1	19.3	15.5	12.1	10.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.5	15.8	17.4	18.2	20.7	21.7	23.8	24.1	23.5	20.4	16.7	15.2
		MCDB	22.2	22.8	27.2	29.8	32.6	34.5	36.1	34.9	35.2	29.1	24.0	20.0
	2%	WB	14.0	14.3	16.0	16.7	19.3	20.7	22.7	22.6	22.3	19.1	15.5	14.0
		MCDB	19.1	21.5	25.1	27.0	29.8	33.0	34.3	34.0	33.2	28.8	23.9	18.5
	5%	WB	13.0	13.3	15.0	15.6	18.1	20.0	21.8	21.7	21.2	18.0	14.6	12.9
		MCDB	18.9	20.4	23.1	25.1	27.7	31.0	33.0	33.3	31.9	27.3	22.5	17.9
	10%	WB	12.1	12.4	14.0	14.7	17.0	19.1	21.0	20.8	20.2	17.0	13.9	11.9
		MCDB	18.3	18.5	21.1	23.1	25.5	29.0	31.5	32.2	30.4	25.2	21.0	16.9
Mean Daily Temperature Range	5% DB	MDBR	13.0	12.6	13.3	13.6	13.4	14.8	15.6	16.2	15.7	14.4	14.0	12.6
		MCDBR	17.2	18.5	18.2	19.1	18.9	18.4	17.2	18.0	18.8	19.7	18.7	17.6
		MCWBR	7.7	7.8	7.5	7.3	7.1	6.2	5.6	6.3	6.3	7.3	7.6	8.2
	5% WB	MCDBR	13.2	14.2	15.5	16.6	16.6	17.0	15.8	16.5	16.6	16.0	14.5	12.6
		MCWBR	6.8	6.7	7.1	6.9	6.6	6.1	5.5	6.0	6.1	6.3	6.7	6.6
Clear-Sky Solar Irradiance	taub		0.303	0.313	0.335	0.349	0.356	0.362	0.383	0.367	0.350	0.340	0.305	0.305
	taud		2.547	2.534	2.450	2.416	2.429	2.423	2.428	2.460	2.494	2.493	2.582	2.576
	Ebn at Noon		915	942	941	937	928	917	896	908	913	896	907	893
	Edh at Noon		81	91	107	116	116	116	115	110	101	93	77	75
All-Sky Solar Radiation	RadAvg		3.24	4.13	5.56	6.75	7.39	7.95	7.65	7.24	6.17	4.70	3.60	2.94
	RadStd		0.22	0.40	0.27	0.30	0.39	0.40	0.36	0.18	0.19	0.28	0.17	0.20

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+1.13	+1.45	N/A	+0.47	N/A	N/A	-227	+435	+246
(47)	Regional (51 neighbors)	+0.63	N/A	-1.70	+0.72	N/A	N/A	-112	+223	+112

Nomenclature: See separate page